SHAFIA TASNIM SAMONTI

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OBJECTIVE

Final-year Industrial and Production Engineering student at BUET with strong analytical, problem-solving, and cross-functional collaboration skills. Experienced in leading diverse teams and delivering impact through real-world projects and extracurricular initiatives. Passionate about digital transformation and eager to contribute to Grameenphone's Nextern Internship Program by applying a learning-by-doing mindset to develop innovative, data-driven solutions.

EDUCATION

Bangladesh University of Engineering & Technology 2022 – Present
Bachelor of Science in Industrial & Production Engineering (Ongoing)
CGPA: 3.64 (Average till 4-1)
Awards: University Stipend at Level-1/Term-2, Level-2/Term-2, Level-3/Term-1, Level-3/Term-2 & Level-4/Term-1

Rajshahi College 2020
Higher Secondary Certificate (HSC)
GPA: 5.00

WORK EXPERIENCE

Industrial Attachment 10 Nov 2025 – 04 Dec 2025
PRAN Industrial Park - PRAN-RFL Group

- Analyzed high-capacity FMCG production systems including **UHT milk (12,000 L/hr), Juice (230,000 packs/day), and Powder Milk (8,000 kg/day)** to understand integrated dairy and beverage operations.
- Evaluated a Kaizen initiative eliminating **40% rejection of 2,000 L/hr returned milk**, projecting **BDT 26.4 crore annual savings** via aseptic tank integration.
- Assessed **HACCP-based CCP & OPRP controls** across pasteurization (83–85°C), UHT sterilization ($\geq$135°C), and metal detection to ensure food safety compliance.
- Reviewed **multi-effect evaporation and spray drying systems** (170°C inlet air) achieving $<$**0.01% production waste**.
- Conducted **throughput, bottleneck, and line balancing analysis** in juice and frozen food packaging; proposed automation reducing manual labor from **75% to 10%** with break-even evaluation.
- Analyzed **CIP (85–90°C), homogenization (120–170 bar), PLC-controlled packaging, and plant utilities (Boiler, Compressor, Chiller, WTP, ETP)** to assess operational reliability and maintenance practices.
- Collaborated cross-functionally (Production, QC, Engineering, Maintenance) on **quality metrics, packaging integrity tests, and continuous improvement initiatives**.

PROJECT

Multi-purpose Plumbing Machine

- Designed a **motorized pipe cutting and threading machine** to improve productivity and safety in construction plumbing operations.
- Conducted a **survey of 45 plumbers**; identified key inefficiencies including **71% high time consumption, 69% excessive physical effort, and 56% precision issues** in manual methods.
- Applied **Quality Function Deployment (QFD)** to translate customer needs into engineering specifications; prioritized portability (80%) and adjustable pipe sizing (67%).
- Modeled full assembly in **SolidWorks** and performed **ANSYS static structural analysis** evaluating stress, deformation, fatigue life, and safety factor of critical components.
- Executed **material selection (Digital Logic Method)**, manufacturing process selection, and **cost, break-even, and sensitivity analysis** to validate technical and economic feasibility.

SKILLS

Programming & Scripting: Python, MATLAB, C, C++
Data Handling & Visualization: Microsoft Excel
Tools & Platforms: SolidWorks, AutoCAD, CATIA, ANSYS, Simulink
Manufacturing & Industrial Engineering: Production Planning, Process Improvement, Line Balancing, Forecasting, Supply Chain Fundamentals
Graphics & Design: Adobe Illustrator, Blender, Canva
Soft Skills: Analytical Thinking, Problem Solving, Leadership, Communication, Cross-functional Collaboration, Time Management, Adaptability

ACHIEVEMENTS

- Champion, 5th Cohort – University Hub Program 2024 (RISE, BUET)
- Champion, Ideation Challenge 5.0 – Orange Corners Bangladesh
- Champion, Idea Presentation – Bangladesh Youth Climate Sustainability & Innovation Summit 2025
- 70% Scholarship Recipient – ISCEA Prize Global Case Competition 2025
- Finalist – Hult Prize 2026
- 1st Place (Preliminary Round) – Anatolian Rover Challenge 2023
- 2nd Place (Preliminary Round) – Anatolian Rover Challenge 2024
- Runners-Up – Video Essay Challenge, BUET Brainiacs

EXTRA-CURRICULAR ACTIVITIES

CCI Next Talent Campus Ambassador /*Coca-Cola Bangladesh* June 2025 – Present
- Attended orientation and seminar led by Supply Chain Head, focusing on strategic thinking.
- Promoted Coca-Cola job openings on campus & assisted students with applications.

Mentee of Womentor 6.0 /*Banglalink* Dec 2024 – July 2025
- One-on-one mentorship by Banglalink's top employee Syaba Trannum Quader focused on career development.
- Completed 12 intensive career sessions at Banglalink Tiger's Den (communication, personal branding).
- Engaged in experience day at SOS Children's Village, fostering empathy and social awareness.

Vice President /*BUET Entrepreneurship Development Club* Apr 2025 – Present
- Arranged events like EDC Talk, Taka Talk, TEDx BUET, BUETian Entrepreneurship Summit 2026.
- Designed creative materials to enhance club branding.
- Hosted events, demonstrating strong communication skills.

Campus Ambassador /*ISCEA Prize Case Competition 2025* June 2025 – Present
- Attended orientation and seminar led by Ejazur Rahman, focusing on strategic thinking.
- Engaged in creative group activities to enhance team collaboration and problem-solving.
- Organized seminar for ISCEA Case Competition for around 220 students.
- Engaged and coordinated with students extensively.

Campus Coordinator /*Orange Corner Bangladesh* Mar 2024 – Apr 2025
- Coordinated workshop, seminar, and event booth at BUET in collaboration with EDC.
- Participated in OCB cohort programs; facilitated career-focused sessions.

Campus Innovative Coordinator /*Needle Innovation Challenge* Jan 2024 – Feb 2025
- Arranged seminar and event booth at BUET.
- Participated in head Office visiting and gaining practical insights.

Design Secretary /*Association of Industrial & Production Engineers (AIPE)* Apr 2025 – Present
- Organized excel competition for IPE Fiesta-2026.
- Organized poster presentation competition for IPE Fiesta-2024.
- Designed promotional materials and managed departmental programs.

Senior Analyst of Science Analysis Sub Team /*Team Interplanetar Mars Rover Team* Apr 2025 – Present
- Contributed to European Rover Challenge (ERC), Anatolian Rover Challenge, and International Rover Design Challenge.
- Played key role in scientific analysis and mission design.

REFERENCES

Dr. Ahsan Akhter Hasin
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